

Overall Comments

I've found the book to be engaging and thought-provoking. It takes a distinctive approach compared to most data visualization texts. In fact, I would describe it less as a traditional visualization book and more as a book about analyzing multiple response variables simultaneously, with visualization as a central tool.

Some particular strengths stood out:

1. **Breadth of R functionality.** The book introduces a range of R packages and features that were new to me, including capabilities within packages I already use (e.g., adding polynomial fits in ggplot2). I particularly appreciated seeing multiple ways to visualize the same dataset.
 2. **Historical context.** The historical notes and quotations are excellent. They provide helpful context for why certain methods are used today. I especially enjoyed the discussion of the development of the general linear model.
 3. **Conceptual clarity.** Many of the central ideas, such as projections and guided tours, are explained clearly and intuitively. While I am unsure how often I would personally use every method presented, I found it valuable to see alternative analytic perspectives. For example, summarizing data with ellipses may not always be ideal, but it seems particularly effective for conveying regression coefficient uncertainty which could be useful for explaining collinearity to students.
 4. **Choice of datasets.** For the most part, the datasets are well chosen. While most datasets in the book fall between real and complete toy datasets, overall they effectively explained the concepts presented. Personally, I would prefer fully real datasets when possible (e.g., say data from IPUMS, etc.) since they provide stronger evidence that the methods presented actually work on real data, the semi-toy datasets used here are valuable for explaining how the methods operate and are fairly interesting.
 5. **Concerns about Chapter 9.** While much of the material was new to me (so I do not claim deep expertise in these methods), I believe there may be some important inaccuracies in the presentation in Chapter 9. Specifically, the null hypothesis stated on p. 276 appears incorrect. The analysis is not testing **FIXED**
 $H_0: \mu_{BM} = \mu_{WP}$,
but rather whether the mean *vectors* are equal across groups (i.e., whether the joint mean vector for (BM, WP) differs between groups).
Moreover, the results do not imply that both BM and WP differ individually; they indicate that the joint vector differs, which could be driven by only one component. This is therefore a different inferential question than conducting separate univariate tests on BM and WP. It is not simply a “more powerful” version of the univariate analysis, but a distinct multivariate test with a different null hypothesis and interpretation.
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Specific Comments

- **p. 19:** Correlation was developed by Galton in the latter half of the 19th century (not the 18th).
- **p. 26:** The text should indicate that the woman mistakenly reported her height (not her weight).
- **p. 29:** It would be helpful to include a confidence interval for the regression slope of *Day*.
- **p. 34:** One might argue that confidence intervals and hypothesis tests could be as informative than the visualizations presented for the draft dataset. (This is how I typically use this dataset in the introductory statistics course I teach.)
- **p. 133 (Fig. 4.16):** The bar plot should preserve sign (e.g., with bars above and below zero). Directionality is important for interpretation. For instance, the fact that *murder* and *auto* have opposite signs on Dim-2 appears central to interpreting that dimension as distinguishing violent from non-violent crimes, but this is obscured if signs are not shown.
- **p. 184:** It is unclear which package provides the `confidenceEllipse()` function.
- **p. 273 (second paragraph):** For clarity, it may be preferable to use y (rather than x) for the collection of response variables (e.g., $y = y_1, y_2, \dots, y_p$).
- **p. 275 (footnote):** Typo: “test that gives.”
Also, I would hesitate to describe a permutation test as “exact” in a strict sense, since it does not generally generate the exact null distribution (though it does control Type I error under weaker assumptions than parametric tests).
- **p. 278 (Fig. 9.2):** It would be informative to include marginal distributions (e.g., KDE plots in the margins) to show how separated the variables are. One could also consider showing a density along the minor axis of the ellipses to illustrate separation along that direction.
- **p. 284 (Figs. 9.5–9.6):** There appears to be noticeable overlap in the projections in Fig. 9.5, yet strong separation in the right-hand plot of Fig. 9.6. It is unclear whether the same projection is being shown in both figures.
- **p. 285:** `pivot_longer()` has replaced `gather()` in `tidyr`; updating the code would be helpful.
- **p. 288:** The output of `summary(Anova)` (e.g., Pillai, Wilks, etc.) would benefit from additional explanation, as readers may not be familiar with the interpretation of these statistics.

Comments on Part IV

Overall Comments

Part IV focuses more heavily on methods developed or popularized by the author, particularly HE plots. While I found these methods interesting, I was not entirely convinced that they consistently provide substantial additional insight beyond more standard approaches.

In particular, the motivation and interpretation of HE plots could be clarified further. As presented, it seems that their primary use is to visualize when we can reject the null hypothesis that group mean vectors are not equal, along with providing some sense of covariance structure (e.g., relationships among residuals and fitted values). If that is the main goal, it would be helpful to state this more explicitly and to articulate more clearly what unique insight HE plots offer relative to other techniques (i.e., is the main value that is it easier to look at these plots than tables of statistics, and if so, I'm not sure it is that big of a win relative to these plots giving clear new insights into the questions of interest).

Relatedly, a more detailed and intuitive explanation of the H and E matrices (e.g., at the top of p. 299) would strengthen this section. Similarly, expanding the discussion of multivariate test statistics - such as Pillai's trace and Wilks' lambda (introduced around p. 302) - would be valuable. These statistics are used repeatedly throughout the text, but the explanation provided does not fully convey what they reveal about the data or how to interpret differences between them in practice.

As in earlier chapters, the datasets fall somewhere between toy examples and real-world data. Most are engaging and help sustain interest, although a few (e.g., the plastic film example) could benefit from clearer exposition. I also appreciated seeing the author's modeling choices (such as contrasts), as these are often omitted in other texts.

That said, I occasionally found the presentation repetitive, particularly in Chapter 10, because at times struggled to see what new insights were being gained from the analyses. This may partly reflect gaps in my own understanding, but it also seemed that the examples did not always demonstrate clear, compelling advantages of the proposed methods.

Chapter 14 stood out as a highlight because it analyzes a real-world dataset, so we can see how the methods can be used on a problem that is not a toy example. However, the insights provided by the methods in the book felt somewhat limited, which reinforced my broader concern about how useful these methods are.

In summary, the book is valuable in that it introduces readers to a set of methods that are not widely covered in modern texts (and may have been more prominent historically). However, the case for their practical utility could be made more convincingly. While many of the explanations were very good (e.g., I thought CCA was explained well), improving the clarity of some of the other key methods and the added value of these methods, especially on rich, real-world datasets, would strengthen the book.

Specific Comments

- **p. 281:** The equation at the top of the page (above the first code block) appears to contain small errors. It should read:
$$p_1 + (d^T(a - p_1) / d^T d) \cdot d, \text{ where } d = p_2 - p_1.$$

As written, the equation is missing the final multiplication by d , and d is defined with the incorrect sign. The code below the equation appears to implement the correct version, so the text and code are currently inconsistent.
- **Figures 9.5–9.6:** In Figure 9.6 (right panel), the projected points show clear separation between the two classes along a direction that appears to maximize class separation. However, in Figure 9.5, projecting onto the line connecting the class means does not

produce the same separation. This suggests that the mean-difference direction does not maximize separation. It would be helpful to clarify whether a different projection is being used in Figure 9.6 (e.g., one that accounts for the residual covariance structure), or whether there is a mismatch in the figures.

- **pp. 288 and 302:** Several multivariate test statistics (e.g., Pillai, Wilks) are introduced on p. 288 but not explained until later. While p. 302 provides some explanation, it would be helpful to expand this discussion to give clearer intuition about what each statistic measures and how they differ in practice. Given their repeated use, a more thorough explanation would significantly improve accessibility.
- **p. 347:** The discussion of Huang (2019), which questions the usefulness of MANOVA, is quite brief. Expanding this discussion, perhaps earlier in the book, would be valuable. Since a central goal is to introduce useful tools, a more explicit treatment of the strengths and limitations of these methods (especially relative to alternatives) would help readers better understand when and why to use them.

Minor Suggestions

- **pp. 283–285 (and elsewhere):** It would be preferable to use `pivot_longer()` instead of `gather()`, as the latter has been deprecated in `tidyr`.
- **p. 372:** Two variable descriptions in the bullet points appear to be swapped. Time counting is positively correlated with reading and math, while visits to school are in the opposite direction.
- **p. 404:** In the expression for $B(-I)$, the term for Y appears to be missing the $-I$ subscript.
- **Figure 14.1:** A beeswarm (or similar) plot may be more effective than overlaying a violin plot with individual data points, which can lead to overplotting.
- **Figure 14.2:** A pairs plot with displayed correlation values and scatterplots may provide more insight than the current visualization, which obscures some of the underlying relationships.